

Welcome to Day 4 of Computational Physics!

This is a Markdown cell. The code here is really just text, but with some *slight modifications*. Markdown is great because you can see exactly what is going to appear and if you do not have a renderer for Markdown its fine because you can still read it. It is used throughout the coding world as the default for any kind of human-readable text. You can use inline equations like $\psi = \int \vec{E} \cdot d\vec{a}$ or you can enter centered equations such as:

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \right] \Psi(x, t)$$

There are many features of markdown that we will see over the semester, but I'll use this as a way to show you what is going on in each day's file. It may sound silly, but this is one of those soft skills that can really impress a future employer, so this is definitely worth paying attention to.

```
In [10]: x = 15

if x<10:
    print(x**2)
elif 25>x>10:
    print(x**0.5)

print('done')
```

3.872983346207417
done

```
In [18]: h = float(input("Enter the height of the tower: "))
t = float(input("Enter the time interval: "))
g = 9.81

y = h - 1/2*g*t**2
if y>0:
    print('The height of the rock is ', y, 'meters')
else:
    print('The height of the rock is 0 meters.')
```

The height of the rock is 2.3359375 meters

```
In [24]: x = 10

while x>1:
    x = x/2
    print(x)
    if x<2:
```

```
        break  
    print('done')
```

```
5.0  
2.5  
1.25  
done
```

In [25]: x

Out[25]: 1.25

```
In [30]: for i in range(0,10, 2):  
         print(i)
```

```
0  
2  
4  
6  
8
```

In [27]: i

Out[27]: 9

In []: